

Curriculum Vitae

PERSONAL DATA

Full name Yujin Choi
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GitHub <https://github.com/uzn36>
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PROFESSIONAL EXPERIENCE

April. 2026 Postdoctoral Researcher
- Current Nanyang Technological University, Singapore
The College of *Computing and Data Science*
Host: Prof. Jaehong Yoon
Mar. 2026 Postdoctoral Researcher
- Current Ulsan National Institute of Science and Technology
Department of *Industrial Engineering*
Mentor: Prof. Saerom Park

EDUCATION

Mar. 2021 Seoul National University, Seoul, Korea
- Feb. 2026 Ph.D. candidate in *Industrial Engineering*
Advisor: Prof. Jaewook Lee
Cumulative GPA 3.84/4.3
Dissertation:
Trustworthy Generative Models: from Privacy to Fairness
Mar. 2017 Yonsei University, Seoul, Korea
- Feb. 2021 Bachelor of Engineering in *Industrial Engineering*
Bachelor of Science in *Mathematics (Double Major)*
Cumulative GPA 3.91/4.3

PUBLICATIONS

[Top Conferences]

[C1] **Choi, Y.***, Park, Y.*, Byun, J., Lee, J., & Park, J. (2025). Safeguarding Privacy of Retrieval Data against Membership Inference Attacks: Is This Query Too Close to Home?. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 8241–8258, Suzhou, China. Association for Computational Linguistics.

[C2] **Choi, Y.***, Park, J.*, Kim, H., Lee, J., & Park, S. (2024, March). Fair sampling in diffusion models through switching mechanism. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38, No. 20, pp. 21995-22003).

- [C3] Park, J.*, **Choi, Y.***, & Lee, J. (2024). In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 12236-12246).
- [C4] Park, J., **Choi, Y.**, & Lee, J. (2025). Multi-Class Support Vector Machine with Differential Privacy., *accepted for Advances in Neural Information Processing Systems*
- [C5] Park, J., Kim, H., **Choi, Y.**, & Lee, J. (2023, July). Differentially private sharpness-aware training. In *International Conference on Machine Learning* (pp. 27204-27224). PMLR.
- [C6] Kim, H., Park, J., **Choi, Y.**, & Lee, J. (2023). Fantastic robustness measures: The secrets of robust generalization. *Advances in Neural Information Processing Systems*, 36, 48793- 48818.
- [C7] Park, J., Lee, S., Jeong, W., **Choi, Y.**, & Lee, J. (2024). Leveraging priors via diffusion bridge for time series generation. Accepted for *KDD 2026*.

*: equal contribution

[Journals]

- [J1] **Choi, Y.**, Park, J., Park, Y., Lee, J., & Byun, J. (2026). Differentially Private Upsampling for Enhanced Anomaly Detection in Imbalanced Data, *Engineering Applications of Artificial Intelligence*. **(IF: 8.0, top 2.8%)**
- [J2] **Choi, Y.**, Kim, D., & Lee, J. (2025). Temporal Consistency Ensemble Empirical Mode Decomposition for forecasting practical metal price, *Engineering Applications of Artificial Intelligence*, 158, 111490. **(IF: 8.0, top 2.8%)**
- [J3] **Choi, Y.**, Park, J., Lee, J., & Kim, H. (2023). Exploring diverse feature extractions for adversarial audio detection. *IEEE Access*, 11, 2351-2360. **(IF: 3.6, top 35.0%)**
- [J4] Byun, J., **Choi, Y.**, & Lee, J. (2025). Improving the utility of differentially private clustering through dynamical processing. *Pattern Recognition*, 157, 110890. **(IF: 7.6, top 9.0%)**
- [J5] Byun, J., **Choi, Y.**, Lee, J., & Park, S. (2024). Privacy-preserving inference resistant to model extraction attacks. *Expert Systems with Applications*, 256, 124830. **(IF: 7.5, top 6.6%)**
- [J6] Park, J., **Choi, Y.**, Byun, J., Lee, J., & Park, S. (2023). Efficient differentially private kernel support vector classifier for multi-class classification. *Information Sciences*, 619, 889-907. **(IF: 6.8, top 8.1%)**
- [J7] Park, J., Kim, H., **Choi, Y.**, Lee, W., & Lee, J. (2023). Fast sharpness-aware training for periodic time series classification and forecasting. *Applied Soft Computing*, 144, 110467. **(IF: 6.6, top 13.7%)**
- [J8] Byun, J., Park, S., **Choi, Y.**, & Lee, J. (2023). Efficient homomorphic encryption framework for privacy-preserving regression. *Applied Intelligence*, 53(9), 10114-10129. **(IF: 3.5, top 41.2%)**

PATENT

- [1] Clustering method and apparatus supporting differential privacy, Application No. 1020220185186, filed 2022.12.27, Registration No. 10-2815746, registered 2025.05.28
- [2] Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector, Application No. 10-2023-0073178, filed 2023-06-07
- [3] Method and system for detecting adversarial speech example, Application No. 1020230087024, filed 2023.07.05

PREPRINT AND UNDER REVIEW PAPERS

- [1] **Choi, Y.**, Park, J., Byun, J., & Lee, J. (2024). Leveraging Programmatically Generated Synthetic Data for Differentially Private Diffusion Training. *arXiv preprint arXiv:2412.09842*.
- [2] **Choi, Y.**, Park, J., Park, Y., & Lee, J. (2025). Multiple Hyperplanes with Dual Encoding Support Vector Machine for Industrial Anomaly Detection, *2nd major revision at IEEE Transactions on Industrial Informatics*.
- [3] Kim, H., Park, J., Lee, W., Lee, J., & **Choi, Y.** (2023). Exploring the Effect of Multi-step Ascent in Sharpness-Aware Minimization. *arXiv:2302.10181*., *submitted to IEEE Access*.
- [4] Park, J., **Choi, Y.**, Park, Y., & Lee, J. (2025). Privacy-Preserving Asymmetric Banach Kernel Encoding for Support Vector Classification, *submitted to Pattern Recognitions*.

- [5] Kim, H., Park, J., **Choi, Y.**, Lee, S., & Lee, J. (2024). BayesNAM: Leveraging Inconsistency for Reliable Explanations. *arXiv preprint arXiv:2411.06367*, submitted to *Pattern Recognitions* .
- [6] Kim, H., Park, J., **Choi, Y.**, & Lee, J. (2023). Stability Analysis of Sharpness-Aware Minimization. *arXiv preprint arXiv:2301.06308*, under review at *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

SCHOLARSHIP AND FELLOWSHIPS

[Scholarship]

- National Science and Technology Scholarship (full tuition), Ministry of Science, Korea, 2019-2020
- Youlchon AI Scholarship, Youlchon AI Star, 2025

[Fellowship]

- Doctoral Student Research Grant, Ministry of Science and ICT, Korea (Graduate Research Fellowship in Science and Engineering), 2023-2025

LECTURES AND TEACHING ASSISTANT EXPERIENCE

[Lecturer]

- Financial Programming 2, Soongsil University (2nd semester, 2024)
- Financial Machine Learning, Soongsil University (1st semester, 2025)

[Teaching Assistant]

- **Industry training programs**
 - Samsung Data Scientist for Device Solution for Prof. Jaewook Lee
 - Optimization and Linear Algebra (2021-2022) - **Best TA Award, 2021**
 - Optimization (2023-2025)
 - Woori Bank: Big data analysis with Python and deep learning with PyTorch (2022)
 - HD Hyundai: Nonlinear Time Series Analysis (2023-2024)
- **National training programs**
 - SNU Big-Data FinTech: Optimization and Machine Learning for Prof. Jaewook Lee (2022-2024)
 - Big Data AI CEO Program (2024-2025)
- **University Courses**
 - Mathematical Methods for Industrial Management, Seoul National University (2nd semester, 2022)